

Reward Shaping and Curriculum PPO for Sparse-Reward SoccerTwos Control

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Abstract: This project trains SoccerTwos agents for a sparse-reward Unity ML-Agents soccer task using Proximal Policy Optimization (PPO). I compare a plain PPO baseline, a reward-shaped PPO variant, and curriculum PPO policies trained with progressively harder start states. The main source-visible reward change adds clipped dense bonuses for reducing player-to-ball and ball-to-goal distances during training. The strongest exported policy, `TEAMNAME_v3_AGENT`, is a curriculum PPO agent trained for 30.0M environment steps and evaluated through the repository’s headless match runner. Curriculum PPO produced the best observed learning reward and beat the baseline PPO package in 8 of 10 recorded evaluation matches.

Keywords: Reinforcement Learning, SoccerTwos, Curriculum Learning

1 Overview

SoccerTwos is a competitive 2-vs-2 soccer environment built on Unity ML-Agents [1] and distributed through a starter kit [2], where goals are sparse and useful exploration is difficult. The implemented training pipeline uses Ray RLlib PPO [3, 4] with PyTorch policies, training a single controlled player in the `team_vs_policy` variation against a fixed opponent policy. The training interface uses a 336-dimensional observation vector and a flattened `Discrete(27)` action space; exported agents convert flat logits back to the live branched action interface when needed.

2 Method Description

PPO baseline. The baseline uses RLlib PPO with a feed-forward ReLU policy/value network and shared value layers. PPO optimizes a clipped surrogate objective, which limits each policy update so rollout data can improve the policy without very large destructive probability-ratio changes. The baseline and shaped variants use one hidden layer of width 512, `vf_share_layers=True`, and the default sparse environment reward.

Reward-shaped PPO. The reward-modified agent wraps the training environment with a dense shaping term computed from the environment `info` dictionary. Let d_{pb} be the player-to-ball distance and d_{bg} be the ball-to-goal distance. At each transition, the wrapper adds

$$r_{\text{shape}} = \text{clip} \left(0.01(d_{pb}^{t-1} - d_{pb}^t) + 0.02(d_{bg}^{t-1} - d_{bg}^t), -0.05, 0.05 \right).$$

This rewards movement toward the ball and ball progress toward the target goal while keeping the bonus small relative to scoring events. The shaping code is used only during training, so the submitted agent does not depend on privileged `info` fields at inference time. This follows the motivation of reward shaping for accelerating sparse-reward learning [5].

Curriculum PPO. The curriculum variant changes the learning setup rather than the final environment API. A callback sets ball and player initial states through the Unity environment

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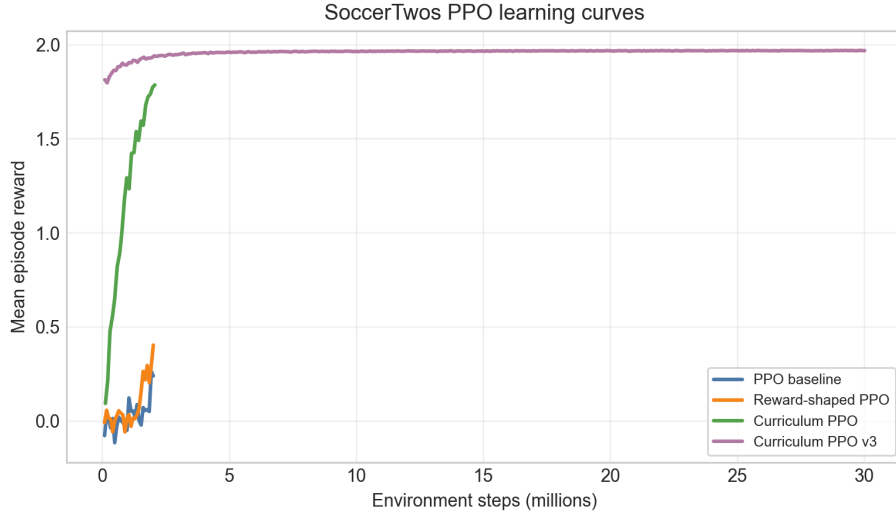


Figure 1: Mean episode reward over training steps for baseline PPO, reward-shaped PPO, curriculum PPO, and the extended curriculum v3 policy.

Table 1: Recorded headless evaluation summaries. Match counts are small and should be interpreted as preliminary evidence.

Matchup	Episodes	Wins	Agent mean	Opponent mean
Curriculum v3 vs. baseline	10	8	1.1140	-1.2084
Curriculum v3 vs. curriculum	10	6	0.3407	-0.4258

channel and advances across five tasks, from “Very Easy Goal” to “Random Players,” when `episode_reward_mean` exceeds 1.5. Curriculum PPO uses a deeper MLP, [256, 256], with ReLU activations and shared value layers. The final v3 policy resumed the curriculum run and extended training to about 30.0M environment steps, following the curriculum-learning hypothesis that easier early states reduce sparse-reward exploration burden before broadening state difficulty [6].

Final hyperparameters. The submitted v3 run used Ray RLlib PPO with PyTorch, learning rate 5×10^{-5} , $\gamma = 0.99$, PPO clip 0.3, 30 SGD epochs, train batch 80,000, minibatch 8,000, rollout fragment 2,000, gradient clip 0.5, 40 workers, 1 GPU, shared value layers, and a ReLU MLP [256, 256].

3 Preliminary Results

Figure 1 compares the learning curves from the available Ray `progress.csv` logs. The baseline PPO run reached final mean reward 0.2399 after 2.0M steps. Reward-shaped PPO improved the same-budget final reward to 0.4036. The first curriculum run reached 1.7869 at 2.06M steps, and the extended v3 run reached final mean reward 1.9692 with a best logged mean reward of 1.9705 at 29.85M steps.

4 Reward, Observation, and Architecture Modifications

Reward modification. The concrete reward change is the training-only `RewardShapingWrapper`. It computes distance deltas from `player_info` and `ball_info`, clips the shaping bonus to ± 0.05 , and adds the bonus to scalar or dictionary rewards. Compared with the baseline, this increased the 2.0M-step training reward from 0.2399 to 0.4036, suggesting faster early progress but not the strongest final policy.

Observation and action handling. I did not find a custom observation-space wrapper or Unity-side observation change in the final code. The implemented interface change is configuration-level:

training uses a single-player 336-dimensional vector with flattened `Discrete(27)` actions, while live matches use branched `MultiDiscrete([3,3,3])` actions. The report therefore does not claim a custom observation rewrite.

Architecture and learning modification. The final policy changes both learning schedule and network shape relative to the baseline. Baseline and shaped PPO use a single 512-unit hidden layer, while curriculum PPO uses a two-layer [256, 256] MLP. More importantly, curriculum training changes the distribution of starting states so the agent first learns near-goal behavior before facing harder ball and player placements. The results support the hypothesis: curriculum training produced substantially higher learning rewards and better recorded match outcomes than baseline PPO.

5 Limitations and Future Work

The recorded evaluation set is small: the strongest comparison against baseline contains 10 episodes, not the planned 100-episode final evaluation. No screenshots or videos were found in the project artifacts, so this report uses learning curves and numerical summaries rather than visual gameplay evidence. Future work should run larger evaluations against random, baseline, and TA agents; compare shaped PPO directly against curriculum v3 in head-to-head matches; and test whether the curriculum policy remains robust under fully self-play or multi-agent training.

6 GitHub Link

TODO_ADD_GITHUB_LINK

References

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